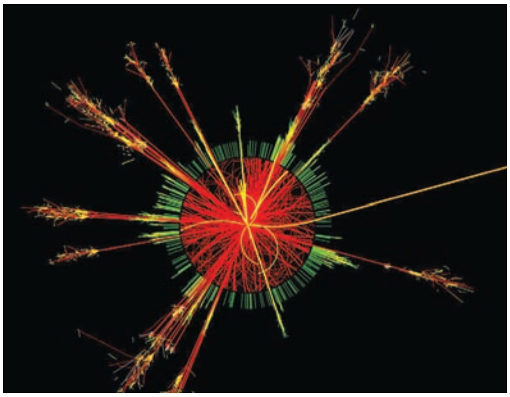

The end of the world as we know it? Simulated production of a black hole in the ATLAS detector, from the CERN web site.

telling people how safe their policies on lending were.

The most worrying argument they have put forward to convince us that the LHC is safe is this: that if tiny BHs are produced in the collider, they would be very short-lived, as they would evaporate by means of Hawking radiation before they can come in contact with any other particles to absorb. This is all very well, but this theory from Stephen Hawking is completely untested, and is right at the edge of our understanding of the universe and how it works. It may very well, therefore, be wrong. The best way to test a theory such as this is not by gambling with the very existence of our planet.

The scientists at CERN certainly seem to expect miniature BHs to be produced in the LHC, and you can find on the CERN web site simulations of the detection of such events, such as the one shown on this page. This is entitled "Simulated production of a black hole in ATLAS". On another of CERN's web pages we find another simulation: "An image of an event in which a microscopic-black-hole was produced in the collision of two protons in a computer generated image of the ATLAS detector." The event is described as an "exciting possibility"! If you could experiment safely with black holes it would indeed be exciting, but if the planet ends up being gobbled up...

On the same page we are reassured with the comment "Such a small black hole would decay instantly to various particles..." Unfortunately, on this and other similar pages, they miss out the all important word "theoretically". It is simply misleading to state as if it were a proven fact that any little BHs would decay instantly. Nobody knows whether this is true or not. Relying on a completely untested theory for the safety of the planet seems to many people to be completely unacceptable.

However, perhaps the best argument in favour of the safety of the LHC is this: ultra-high energy cosmic rays have been hitting our upper atmosphere since the planet was formed. Their collisions with atoms have energies far higher than will be achieved in the LHC. However, some have pointed out that, unlike in the LHC, these cosmic rays will be hitting stationary atoms, and any BHs produced will have high velocities and fly off away from the Earth. However, this will have been happening on the surfaces of every star, planet, neutron star, etc., out there, and we have detected no problems as a result of miniature BHs. For example, if a cosmic ray hit a neutron star and created a persistent black hole (not short lived as in Hawking's theory), this would very likely be captured by the very strong gravitational field of the neutron star and would rapidly consume the star. However, we are able to detect neutron stars, and we can therefore deduce that this is not happening.

We can probably draw one of three conclusions: 1) such collisions do not produce miniature BHs, and if they do, 2) either Stephen Hawking is correct, or if they last a long time, 3) they interact so slowly with normal matter that they remain microscopic and harmless for a period of time at least similar in magnitude to the age of universe, and that they are therefore everywhere — passing through you right now, perhaps. ■

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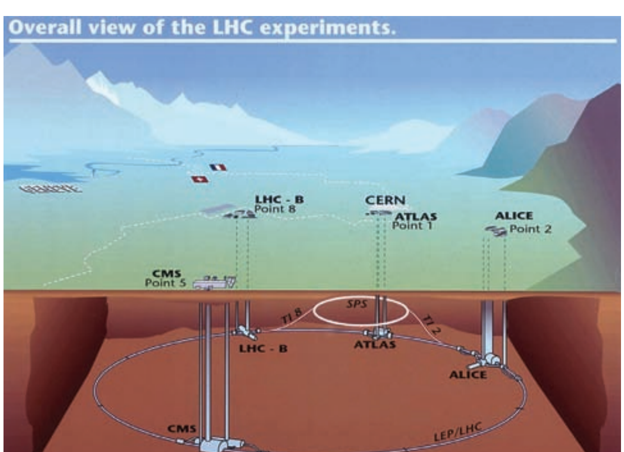

An overall schematic view of the LHC experiment at CERN. This shows the locations of the four main experiments (ALICE, ATLAS, CMS and LHCb). ©CERN